Apart from flybys, there have been only two fully successful missions to Mars: *Mariner 9* in 1971, and *Vikings 1* and *2* in 1976. They revealed a deep rift valley that would stretch from New York to San Francisco; immense volcanic mountains, the largest of which towers 80,000 feet above the average altitude of the Martian surface, almost three times the height of Mount Everest; an intricate layered structure in and among the polar ices, resembling a pile of discarded poker chips, and probably a record of past climatic change; bright and dark streaks painted down on the surface by windblown dust, providing high-speed wind maps of Mars over the past decades and centuries; vast globe-girdling dust storms; and enigmatic surface features.

Hundreds of sinuous channels and valley networks dating back several billion years can be found, mainly in the cratered southern highlands. They suggest a previous epoch of more benign and Earthlike conditions—very different from what we find beneath the tenuous and frigid atmosphere of our time. Some ancient channels seem to have been carved by rainfall, some by underground sapping and collapse, and some by great floods that gushed up out of the ground. Rivers were pouring into and filling great thousand-kilometer-diameter impact basins that today are dry as dust. Waterfalls dwarfing any on Earth today cascaded into the lakes of ancient Mars. Vast oceans, hundreds of meters, perhaps even a kilometer, deep may have gently lapped shorelines barely discernible today. *That* would have been a world to explore. We are four billion years late.*

On Earth in just the same period, the first microorganisms arose and evolved. Life on Earth is intimately connected, for the most basic chemical reasons, with liquid water. We humans are ourselves *made* of some three-quarters water. The same sorts of organic molecules that fell out of the sky and were generated in the air and seas of ancient Earth, should also have accumulated on ancient Mars. Is it plausible that life quickly came to be in the waters of early Earth, but was somehow restrained and inhibited in the waters of early Mars? Or might the Martian seas have been filled with life